Week 3 homework

Python

(and in java if you want)

- Alpacas: https://dmoj.ca/problem/aac1p2
  - In short:
    You enter your alpaca into race, the race track is of length **D**.
    You are competing against **N** number of other alpacas.
    But oh no! your alpacas is pregnant or something idfk, it runs slower now
    Anyways, you decide to cheat with cool ray gun that slows down the other alpacas:
    Each alpacas has their own speed, you can shoot an alpacas with your ~~ray~~ gun. Your ~~ray~~ gun sets the shot alpacas speed by $speed * \frac{100-x}{100}$ and you can only use this ~~ray~~ gun **K** amount of times.
    Given **N**, **D**, **X** and **K** entered by the user, determine if your alpaca can win the race.
  - Example:
    <input>: 2 12 3 30
    #> 2 = number of opponent alpacas, #> 12 = length of race track, #> 3 = amount of times your ~~ray~~ gun can be used #> 30 = effectiveness of your ~~ray~~ gun (plug it into the formula)
    <input>: 100
    #> speed of the first opponent alpaca
    <input>: 50
    #> speed of the second opponent alpaca
    <input>: 50
    #> speed of your own alpaca
    <output>: YES
- Shirts: https://dmoj.ca/problem/ecoo19r1p1
  - In short:
    person starts with **X** amount of shirts, this person does laundry at the start of the day only when all his clean shirts are dirty, person wears **1** clean shirts per day.
    Given **N** amount of days, **M** amount of events where each event will provide person with a new clean shirt, where the events will happen in the next **D** days.
    Provided with on which days each event will happen
    The program will figure out on which days the person will wash his shirts
  - Example:
    <input>: 1 2 10

#> 1 = number of days, #> 2 = amount of events, #> 10 = next days amount

<input>: 2 5

#> there will be an event on day 2 and day 5

<output>: 3